GRAFT: Graph-Regularized Prototype Fields for Realistic Test-Time Adaptation of VLMs

Anonymous CVPR submission

Abstract

Unlabeled test data can improve the zero-shot classifier of a vision-language model, yet transductive and test-time adaptation methods often fall below the zero-shot baseline when only a few vocabulary classes are present in a batch or when streams are correlated. We trace this failure to a shared design: every class owns independent parameters estimated from its own samples, so absent or rare classes are left at biased text anchors or fitted to foreign samples. We propose GRAFT, which models the adapted prototypes as text anchors plus a displacement field that is partially pooled over a semantic class graph through a Gaussian Markov random field prior, together with an explicit class-presence model. Inference is closed form and gradient free on cached features; anchor-only shrinkage and a single global shift are the two extreme settings of the prior, absent classes are provably harmonic extensions of the present ones, and the pooling strength can be chosen by empirical Bayes. Under the realistic batch and online protocols on eleven benchmarks and four ImageNet shifts, GRAFT improves over the zero-shot model in every setting and over anchor-based transduction where few classes are present.

1. Introduction

Vision-language models (VLMs) such as CLIP [24] and SigLIP [38] classify an image by comparing its embedding with the text embeddings of class names. The resulting zero-shot classifier is general but frozen, and a growing family of methods adapts it from the unlabeled test data itself, transductively on a batch [16, 19, 22, 36] or online on a stream [7, 18, 39, 41], without training and on cached features.

Problem. These methods are evaluated on batches that contain every class of the vocabulary in comparable numbers. Realistic batches do not: a photo album or a user session contains a handful of the classes that the vocabulary names, and streams are temporally correlated. Under such conditions transductive methods lose tens of accuracy points with respect to the zero-shot model they are meant to improve [37], and a controlled benchmark of test-time adaptation reports marginal gains, poor transfer to backbones other than CLIP and degraded calibration [29]. The remedy proposed in [37] and refined in [14] is a statistical anchor: each class prototype is shrunk towards its text embedding with a strength that grows with the number of test samples the class receives. Anchoring prevents collapse but treats every class in isolation: a class that receives no samples stays at its text anchor, although that anchor is displaced from the image distribution by the modality gap and by the domain of the test data [20, 28], and a class with one or two samples is adapted from those samples alone. The problem addressed here is therefore to adapt *all* prototypes of a large vocabulary from an unlabeled batch in which most classes are absent or rare, without collapsing on the absent ones and without discarding what the present ones reveal about them.

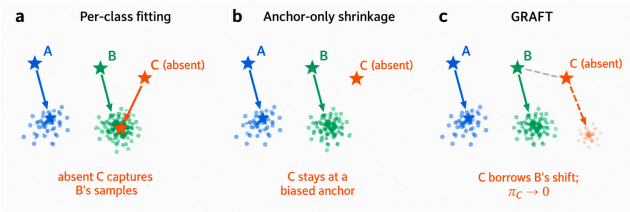


Figure 1. Per-class adaptation on a realistic batch (conceptual illustration). Classes A and B are present, C is absent, and all images are shifted from their anchors. (a) Per-class fitting lets C capture samples of B. (b) Anchor-only shrinkage leaves C at its biased anchor. (c) GRAFT lets C inherit B's displacement and removes it through the presence model.

Approach. The displacement between the text anchor of a class and the centroid of its images contains a component shared by all classes, the modality and domain offset, and a component that varies smoothly with the semantics of the class: breeds of dogs move together, textures move together. GRAFT turns this structure into a prior. The adapted prototypes are written as $\mu_k = t_k + d_k$ and the displacement field $D = [d_1; \dots; d_K]$ receives a Gaussian Markov random field prior whose precision is $\alpha I + \lambda L_T$, where L_T is the Laplacian of a nearest-neighbour graph over the text em-

beddings. The α term reproduces the anchor-only shrinkage of [14, 37]; the λ term couples the classes, so that the displacement of a well-observed class propagates to its semantic neighbours. The maximum a posteriori field has a closed form, $D = (N_z + s(\alpha I + \lambda L_T))^{-1} N_z E$ with soft counts N_z and empirical displacements E: a graph filter that returns the empirical displacement for well-observed classes and the harmonic extension of the neighbours' displacements for absent ones (Fig. 1c). Classes that receive no samples are additionally removed from the competition by a Dirichlet-regularised estimate of the class proportions, which treats the vocabulary as a label-shift problem [8, 27]. All updates are closed form on cached features, and the prior strengths can be selected by maximising the marginal likelihood of the unlabeled batch, which removes the dataset-specific tuning criticised in [29].

Contributions.

- We formulate transductive VLM adaptation as the estimation of a displacement field over the class vocabulary with a graph-structured prior, and derive closed-form block-coordinate inference for batches and streams (Sec. 4); anchor-only shrinkage and a single global shift are its two extreme settings, and absent classes receive the harmonic extension of the present-class displacements.
- We combine the field with an explicit class-presence model and an empirical Bayes rule for the prior strength, so that the vocabulary is treated as a label-shift problem and no pooling strength has to be tuned per dataset.
- We evaluate GRAFT under the realistic batch and online protocols on eleven benchmarks and four ImageNet shifts, with architecture-matched controls that isolate the graph coupling and the presence model (Sec. 6), and release a unit-tested implementation that plugs into the official harness.

2. Related Work

Unlabeled adaptation of vision-language models. Transductive methods classify an unlabeled batch jointly: TransCLIP [36] fits a Gaussian mixture regularised by a KL term to the zero-shot posterior and a sample-graph Laplacian, EM-Dirichlet [22] works on the simplex, ZLaP [16] and ECALP [19] propagate labels over image and prompt graphs, and LATA [1] and Conf-OT [31] use transduction for conformal prediction. Online methods adapt over a stream with feature caches (TDA [18], DMN [41], BoostAdapter [40]), evolving prototypes (DPE [39]), per-class Gaussians (OGA [7]) or online label distributions (OnZeta [23], LSA [10]); episodic methods act on single images [6, 30, 32, 35]. The realistic protocols with a variable number of effective classes and correlated streams were introduced in [37], which showed that these methods collapse under them and proposed StatA, a shrinkage of each class's Gaussian parameters towards its text embedding with strength $n_k/(n_k + \alpha)$; MOON [14] derives this shrinkage from penalised likelihood on the hypersphere and makes it confidence dependent. Both anchor every class to its own text embedding and use no information from other classes; Sec. 4.2 shows that their mean update is the exact $\lambda = 0$ case of GRAFT. A unified benchmark of test-time adaptation further reports marginal gains, poor transfer to SigLIP and worse calibration [29].

Structure over classes and label shift. Class graphs propagate classifier weights to unseen classes in zero-shot learning [17], propagate prototypes along a category hierarchy [21] and select correlated classes in CLIP test-time adaptation [13]; Laplacian regularisation over samples is standard in transductive few-shot learning [42, 45]. We are not aware of prior work that regularises the *displacement* of VLM prototypes from their text anchors with a Gaussian Markov random field over the class graph; the modality gap literature [20, 28] supports the premise that much of the displacement is shared. Adjusting a classifier to new class priors from unlabeled data is classical [8, 27]; Frolic [44] corrects the label bias of CLIP without labels, and open-vocabulary classifiers degrade as distractor classes are added [2]. Our presence model estimates the proportions inside the same objective as the prototypes.

3. Preliminaries

Zero-shot classification. A VLM provides an image encoder f_v and a text encoder f_t mapping into a common space $\mathbb{R}^d$. For a vocabulary of K class names $\{c_k\}_{k=1}^K$, prompt ensembling yields unit-norm text anchors $t_k = f_t(c_k)/\|f_t(c_k)\|$, collected in $T \in \mathbb{R}^{K\times d}$. A test image I_i gives $x_i = f_v(I_i)/\|f_v(I_i)\|$, collected in $X \in \mathbb{R}^{N\times d}$. The zero-shot posterior is

$$\hat{y}_{ik} = \frac{\exp(\tau\, x_i^\top t_k)}{\sum_l \exp(\tau\, x_i^\top t_l)}, \tag{1}$$

with logit scale τ (100 for CLIP). We write Δ^K for the probability simplex and $\mathbf{1}$ for the all-ones vector.

Unlabeled adaptation. Given X and T only, the goal is to output posteriors $z_i \in \Delta^K$ more accurate and at least as well calibrated as $\hat{y}_i$, for a batch processed at once or for a stream of batches predicted as they arrive; following [37], a batch contains $K_{\text{eff}} \leq K$ effective classes, often far fewer than the vocabulary, and streams are correlated.

Mixture-model transduction and its anchors. TransCLIP [36] and StatA [37] model x_i with a K-component

Gaussian mixture and minimise, over assignments z and parameters, a penalised negative log-likelihood with a KL anchor to $\hat{y}$ and a Laplacian term over a nearest-neighbour graph of the samples. StatA additionally shrinks the mean of class k towards its anchor,

$$\mu_k = \beta_k\,\bar{x}_k + (1-\beta_k)\,t_k, \qquad \beta_k = \frac{n_k}{n_k+\alpha}, \tag{2}$$

with $n_k = \sum_i z_{ik}$ the soft count and $\bar{x}_k$ the responsibility-weighted mean. Equation (2) is the posterior mean of μ_k under an independent Gaussian prior centred at t_k with precision proportional to α. Hence (i) a class with $n_k = 0$ keeps $\mu_k = t_k$ even when every present class has moved by a common offset, (ii) a class with small n_k is estimated from its own few samples only, and (iii) the mixture weights are implicitly uniform, so absent classes keep competing for samples. GRAFT addresses (i) and (ii) by coupling the classes through a graph prior and (iii) by estimating the class proportions.

4. Method

GRAFT keeps the generative view of transduction [36, 37] but changes the object that is adapted: instead of K independent prototypes it estimates a *displacement field* $D \in \mathbb{R}^{K\times d}$ anchored at the text embeddings, $\mu_k = t_k + d_k$, under a prior that couples semantically related classes, together with the class proportions $\pi \in \Delta^K$ and a shared covariance. Figure 2 shows the components: the frozen encoders with the zero-shot prior of Eq. (1) and the two graphs (Sec. 4.1), the likelihood and the graph-regularised field (Sec. 4.2), the class-presence model (Sec. 4.3), the objective (Sec. 4.4), the closed-form inference (Sec. 4.5) and the empirical Bayes rule for the prior strength (Sec. 4.6).

4.1. Sample graph and semantic class graph

Two graphs enter the model. The *sample graph* $W_s \in \mathbb{R}^{N\times N}$, inherited from [36, 37], connects each image to its m_s nearest neighbours in feature space with weight $(W_s)_{ii'} = x_i^\top x_{i'}$; it encodes the manifold assumption that neighbouring images share a label. The *semantic class graph*, new to this work, is built from the text anchors only. Let $S = TT^\top$ be the cosine similarity between anchors. Each class is connected to its m most similar classes with weight

$$w_{kl} = \exp\big((S_{kl}-1)/\rho\big), \tag{3}$$

the neighbourhoods are symmetrised by union, and the normalised Laplacian is $L_T = I - \mathrm{D}^{-1/2} W \mathrm{D}^{-1/2}$ with $\mathrm{D} = \mathrm{diag}(W\mathbf{1})$. $L_T \in \mathbb{R}^{K\times K}$ is positive semi-definite with spectrum in $[0,2]$, is computed once per vocabulary and requires no image; it encodes the hypothesis, stated below, that semantically related classes are displaced from their anchors in similar directions.

4.2. Graph-regularised displacement field

Likelihood. Each class is a Gaussian with mean $\mu_k = t_k + d_k$ and a diagonal covariance $\Sigma = \mathrm{diag}(\sigma^2)$ shared across classes. As in TransCLIP, the likelihood is tempered by κ, which acts as a covariance inflation and keeps the responsibilities soft on d-dimensional features:

$$\ell_{ik}(D,\sigma) = -\frac{1}{2\kappa}\Big[\sum_{j=1}^{d}\frac{(x_{ij}-t_{kj}-d_{kj})^2}{\sigma_j^2} + \sum_{j=1}^{d}\log\sigma_j^2\Big]. \tag{4}$$

Given responsibilities z, the shared variance has the closed-form M-step $\sigma_j^2 = \frac{1}{N}\sum_{i,k} z_{ik}(x_{ij}-\mu_{kj})^2$.

Prior. The field receives a Gaussian Markov random field prior [26] that factorises over feature dimensions,

$$p(D) \propto \exp\Big(-\tfrac{1}{2}\sum_{j=1}^{d} d_{:j}^\top Q\, d_{:j}\Big), \qquad Q = \frac{\alpha I_K + \lambda L_T}{\kappa\,\bar{\sigma}^2}, \tag{5}$$

with $\bar{\sigma}^2 = \frac{1}{d}\sum_j \sigma_j^2$. The term αI_K pulls every displacement towards zero, that is, every prototype towards its text anchor; the term λL_T penalises the Dirichlet energy $\sum_{k\sim l} w_{kl}\|d_k/\sqrt{\mathrm{D}_k} - d_l/\sqrt{\mathrm{D}_l}\|^2$ and therefore asks the displacements of neighbouring classes to agree. Dividing by $\kappa\bar{\sigma}^2$ expresses α and λ in units of sample counts: a prior strength of $\alpha = 1$ is worth one tempered observation.

Closed-form field. Writing $N_z = \mathrm{diag}(n_1,\dots,n_K)$ with $n_k = \sum_i z_{ik}$, $\bar{x}_k = \sum_i z_{ik}x_i/n_k$ and $e_k = \bar{x}_k - t_k$ (with $e_k = 0$ when $n_k = 0$), the maximum a posteriori field given the responsibilities is obtained for every feature dimension j from a $K\times K$ linear system,

$$d_{:j} = \big(N_z + s_j(\alpha I_K + \lambda L_T)\big)^{-1} N_z\, e_{:j}, \qquad s_j = \sigma_j^2/\bar{\sigma}^2. \tag{6}$$

Equation (6) is a graph filter applied to the empirical displacements, weighted by the evidence of each class (Fig. 3): classes with $n_k \gg \alpha+\lambda$ keep their empirical displacement, and classes with little or no evidence receive a smoothed combination of their neighbours' displacements. The d systems share the matrix up to s_j and are solved jointly by preconditioned conjugate gradients.

Special cases. With $\lambda = 0$, Eq. (6) reduces per dimension to $d_{kj} = \frac{n_k}{n_k+\alpha s_j}\,e_{kj}$, the count-based anchor shrinkage of StatA (Eq. (2)) and MOON [14], which are therefore the graph-free member of the family. With $\alpha = 0$, L_T the Laplacian of the complete graph and $\lambda \to \infty$, every class receives the count-weighted mean displacement $d = \sum_k n_k e_k / \sum_k n_k$, a single global shift of all prototypes. GRAFT interpolates between these extremes along the semantic neighbourhoods.

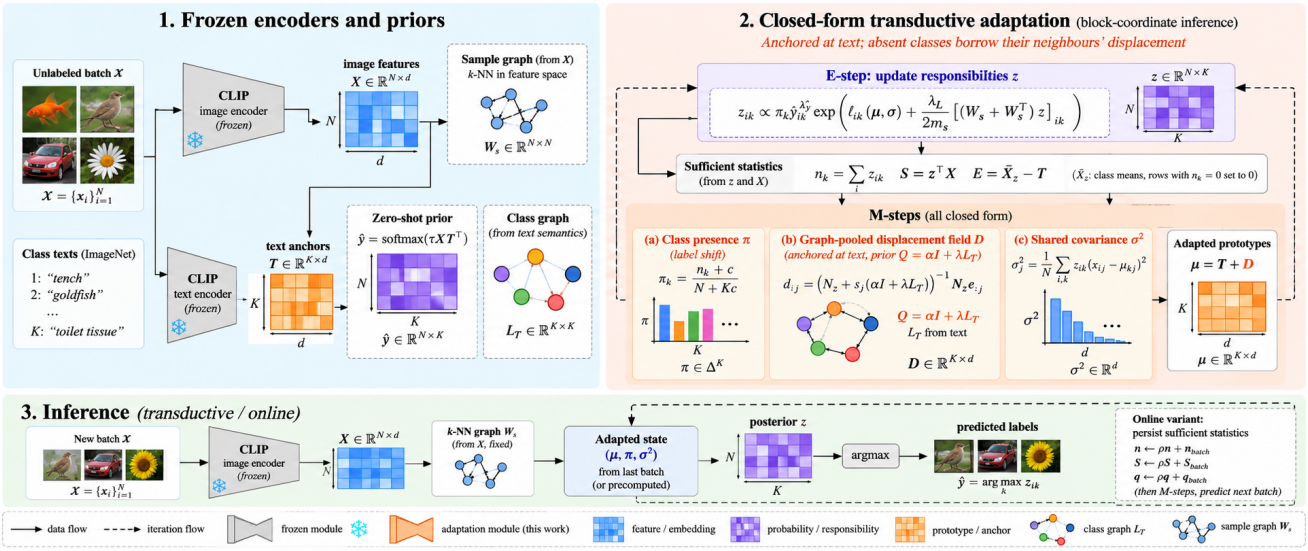


Figure 2. Overview of GRAFT. Frozen encoders give features X and anchors T. The anchors define the zero-shot prior $\hat{y}$ and the class graph L_T, the features define the sample graph W_s. Closed-form block updates alternate the E-step over z with M-steps for the presence π, the graph-pooled displacement field D and the shared covariance; $\mu = T + D$ are the adapted prototypes and z the prediction.

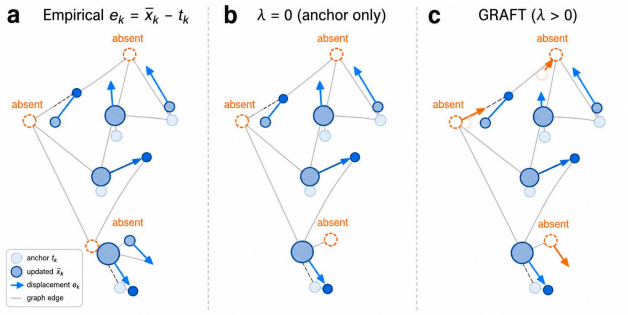


Figure 3. Graph-pooled displacement field on a toy class graph (node size: n_k). (a) Empirical displacements. (b) Independent shrinkage ($\lambda = 0$) leaves absent classes at their anchors. (c) Eq. (6) transfers displacements along the graph.

Absent classes. Let $A = \{k : n_k = 0\}$ be the absent classes, P their complement and $Q_\lambda = \alpha I + \lambda L_T$. Since the rows of Eq. (6) in A have $n_k = 0$ and $e_k = 0$, the solution satisfies $Q_{\lambda,AA}\, D_A + Q_{\lambda,AP}\, D_P = 0$, so

$$D_A = -Q_{\lambda,AA}^{-1} Q_{\lambda,AP}\, D_P \tag{7}$$

is the harmonic extension of the present-class field, and D_P solves Eq. (6) restricted to P with Q_λ replaced by its Kron reduction $Q_\lambda / Q_{\lambda,AA}$ [5]. Absent classes therefore affect the present-class prototypes only through the effective graph, never through data, and are placed where their neighbours' displacements predict (Fig. 1c).

Posterior variance. Given z, the posterior covariance of $d_{:j}$ is $\kappa\sigma_j^2 \left(N_z + s_j Q_\lambda\right)^{-1}$. Since $\lambda L_T \succeq 0$, $\mathrm{Var}(d_{kj}) \leq \kappa\sigma_j^2/(n_k + s_j\alpha)$ for every class k, the variance under independent shrinkage, with strict inequality for every class connected to a present class: pooling never increases the uncertainty of a prototype, and for absent classes it replaces the prior variance by a value controlled by the neighbours' counts. Proofs of the three statements are in the supplementary material; each is also verified numerically by the released unit tests.

Smoothness hypothesis. None of these statements guarantees higher accuracy than zero-shot inference; the benefit of the coupling rests on a testable hypothesis, class-level smoothness (H1): the oracle displacement field $e_k^\star = \mathbb{E}[x \mid y = k] - t_k$ has a Dirichlet energy on L_T substantially smaller than on a degree-matched random graph. If H1 fails, GRAFT reduces in expectation to anchor-only shrinkage; Sec. 7 measures H1 with labels.

4.3. Class presence as label shift

The mixture weights π are free parameters with a symmetric Dirichlet prior $\mathrm{Dir}(c+1)$. Their M-step is the smoothed count

$$\pi_k = \frac{n_k + c}{N + Kc}, \tag{8}$$

and they enter the E-step as $\log \pi_k$, which is the classical label-shift correction of a calibrated classifier [27]: the zero-shot posterior implicitly assumes uniform class priors, and Eq. (8) replaces them by the estimate obtained from the batch. For a batch with $K_{\text{eff}} \ll K$ classes, absent classes receive $\pi_k \approx c/(N + Kc)$ while present ones receive $n_k/(N + Kc)$: with $c = \frac{1}{2}$, a present class with six-

teen samples is more than thirty times more likely a priori than an absent one in a batch of 64 over 1000 classes. The pseudo-count keeps $\log \pi_k$ finite and lets a class absent from earlier online batches re-enter the stream.

4.4. Objective

Collecting the terms, GRAFT minimises over (z, D, σ, π)

$$J = \sum_{i,k} z_{ik}\big[\log z_{ik} - \log \pi_k - \lambda_y \log \hat{y}_{ik} - \ell_{ik}(D,\sigma)\big] - \frac{\lambda_L}{2m_s}\sum_{i,i'}(W_s)_{ii'}\, z_i^\top z_{i'} + \frac{1}{2}\sum_j d_{:j}^\top Q\, d_{:j} - c\sum_k \log \pi_k, \tag{9}$$

where the first line is the variational free energy of the mixture with the text prior $\hat{y}^{\lambda_y}$ of [36], and the second line contains the inherited sample-graph term, the field prior of Eq. (5) and the presence prior. With $\lambda = 0$ and without the π terms this is the StatA objective up to its per-class covariances; with $\alpha = \lambda_L = 0$ in addition, the unanchored mixture.

4.5. Inference

The objective is minimised by block-coordinate descent over four blocks (pseudocode in the supplementary material). The E-step is the mean-field update with the Laplacian term linearised at the current z, a majorise-minimise step [15, 36, 45]:

$$z_{ik} \propto \pi_k\, \hat{y}_{ik}^{\lambda_y} \exp\Big(\ell_{ik} + \frac{\lambda_L}{2m_s}\big[(W_s + W_s^\top)z\big]_{ik}\Big), \tag{10}$$

iterated a few times. The M-steps are the presence update of Eq. (8), the field of Eq. (6) and the shared variance of Sec. 4.2. Each block update is the exact minimiser of Eq. (9) over its block when $\lambda_L = 0$, so J decreases monotonically and converges; with $\lambda_L > 0$ the E-step minimises a majoriser of J, as in [36]. All quantities depend on the data only through the sufficient statistics $n = z^\top \mathbf{1}$, $S = z^\top X$ and $q = \sum_i x_i^{\odot 2}$, which makes the online variant immediate: the statistics are accumulated over the stream with an optional forgetting factor ϱ, and the same M-steps are applied after every batch. The prediction for a batch is its posterior z after adaptation, as in the transductive protocol of [37]; a strictly inductive variant predicts before the statistics are updated. Per outer iteration the E-step costs $O(NKd + Nm_sK)$ and the field $O(n_{\mathrm{cg}}\, Kd\,(1+m))$ with conjugate gradients; no backward pass is executed and the VLM is untouched after feature extraction.

4.6. Empirical Bayes selection of the pooling strength

Fixed (α, λ) are used for all datasets in the main protocol, but the model also admits a label-free rule: under the model, the empirical displacements of the present classes are Gaussian with a covariance that depends on (α, λ) through the Schur complement of Q onto the present classes, and their marginal likelihood is maximised over a small logarithmic grid at every M-step whenever at least three classes are present. The derivation is given in the supplementary material.

5. Experiments

5.1. Datasets

We follow the eleven-dataset suite of TransCLIP and StatA [36, 37] with the CoOp test splits [43]: ImageNet [4], SUN397, FGVC-Aircraft, EuroSAT, Stanford Cars, Food101, Oxford Pets, Flowers102, Caltech101, DTD and UCF101, plus the ImageNet distribution shifts ImageNet-V2 [25], ImageNet-Sketch [34], ImageNet-A [12] and ImageNet-R [11]. Class names and prompt templates are those of CoOp and StatA. Features are extracted once per backbone and cached; no training split is used. Batch adaptation follows the seven settings of [37]: a task draws K_{eff} classes uniformly from $[1, 4]$, $[2, 10]$ or $[5, 25]$ (very low, low, medium) with B=64 images, from $[5, 25]$, $[25, 50]$ or $[50, 100]$ (medium, high, very high) with B=1000, or uses the full test set, with 1000 tasks per dataset. Online adaptation streams batches of 128 whose class composition follows Dirichlet(γ) slot proportions, $\gamma \in \{0.1, 0.01, 0.001\}$ (low to high correlation) or one class after another (separate), with 100 runs per dataset.

The vocabulary sizes range from 10 (EuroSAT) to 1000 (ImageNet) and the test sets from 1,692 (DTD) to 50,000 images (ImageNet); the full list with split sizes is in the supplementary material.

5.2. Baselines

We compare with zero-shot CLIP, the episodic MTA [35], the transductive EM-Dirichlet [22], ZLaP [16], TransCLIP [36], StatA [37] and MOON [14], and the online TENT [33], TDA [18], DMN [41] and OGA [7], all run with official code and default hyper-parameters through the StatA harness into which GRAFT is plugged as a drop-in solver, so that samplers, features and prompts are identical. To attribute gains to the proposed mechanism rather than to the likelihood or the inherited terms, four architecture-matched controls share our implementation: anchor-only shrinkage (λ=0, the StatA/MOON mechanism in our likelihood), complete pooling (a single global shift: α=0, complete graph, $\lambda \to \infty$), presence only (π without displacement) and the unanchored mixture (α=λ=0, no π).

5.3. Implementation details

Inherited constants follow TransCLIP and StatA and are never tuned: κ=50, λ_y=1, λ_L=1, m_s=3, $T_{\max}$=10 outer

iterations with 3 inner E-step sweeps. The new constants are fixed once for all datasets and backbones: α=1 (one pseudo-count, as in StatA), λ=4, m=5 neighbours, ρ=0.05, c=$\frac{1}{2}$. The linear systems are solved exactly for $K \leq 256$ and by 30 preconditioned conjugate-gradient iterations otherwise. Backbones: CLIP ViT-B/16 (main), ViT-L/14 and RN50 [24], OpenCLIP ViT-B/16 trained on DataComp [3], and SigLIP ViT-B/16 [38], the backbone on which [29] found most TTA methods to fail. All runs use a single GPU; seeds and task indices are fixed and shared across methods. We report top-1 accuracy averaged over tasks and datasets, the worst-decile accuracy [7], expected calibration error [9] and time per task.

6. Results

6.1. Realistic batches

Table 1 and Fig. 4 compare GRAFT with the published methods and the architecture-matched controls across the seven batch settings. Three regimes are visible. First, the unanchored transductive methods collapse as the number of effective classes shrinks: TransCLIP falls from 70.3 with all classes to 38.9 in the very-low setting and ZLaP from 66.4 to 27.5, whereas EM-Dirichlet, the strongest method at B=64 (70.3 in the low setting), collapses in the opposite direction, to 33.6 at B=1000 with many classes and 29.5 on the full set. The unanchored control of our implementation reproduces the first pattern (42.4 to 45.4 at B=64). Second, anchoring removes the collapse but adds little beyond it: StatA, MOON and the anchor-only control lie within 1.2 points of each other in every setting (70.4, 71.3 and 70.1 in the very-low setting), so the count-dependent shrinkage that the three share is what prevents collapse. Third, pooling the field across classes adds a margin on top of anchoring that is largest where absent and rare classes dominate and vanishes where every class is well observed: GRAFT improves over the anchor-only control by 1.9, 1.6 and 1.1 points in the very-low, low and medium settings with B=64, by 0.9, 0.7 and 0.5 points with B=1000, and by 0.3 points on the full set, where the special-case analysis of Sec. 4.2 predicts that the two mechanisms coincide; against MOON the margins are 0.7 to 0.1 points in the same order. The presence model alone reaches 67.4 in the very-low setting, about a third of the gain over zero-shot, and the complete-graph control, a single shift of all prototypes, stays below the anchor-only control in every setting (69.6 against 70.1), so the gain comes from pooling along the semantic neighbourhoods rather than from a global shift. Empirical Bayes selection of (α, λ) matches the fixed strengths within 0.1 point, and standard deviations over tasks are 0.2 to 0.4 points for the anchored methods, well below these margins.

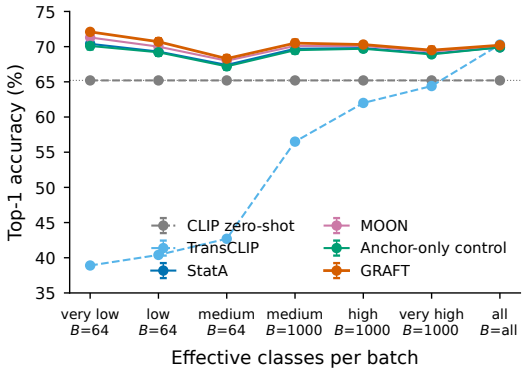


Figure 4. Top-1 accuracy versus the number of effective classes per batch (eleven-dataset mean, error bars: standard deviation over tasks); the dotted line is the zero-shot model.

6.2. Online streams

Table 2 reports the online protocol. The cache-based TDA and DMN are strongest on weakly correlated streams (67.7 and 67.2 at γ=0.1) but gain little on strongly correlated ones (66.8 and 66.3 at γ=0.001), whereas the anchored OGA, StatA and MOON improve with correlation (69.6, 69.5 and 69.9 at γ=0.001), since correlated batches contain few classes with many samples each; TENT never exceeds 65.8. GRAFT follows the trend of the anchored methods and adds 1.2, 1.3 and 1.1 points over StatA at medium and high correlation and on the separate stream, and 0.6 at low correlation, where TDA leads by 0.1; over MOON the margins are 0.2 to 0.9 points. The inductive variant, which predicts each batch before updating the statistics, retains most of the margin (69.5, 70.3 and 69.8 against 70.1, 70.8 and 70.2), so the gain does not depend on transduction within the batch.

6.3. Per-dataset results and distribution shifts

Table 3 reports every baseline on each dataset in the low setting and Fig. 5 plots the zero-shot model, StatA and GRAFT. GRAFT is first on eight of the eleven datasets and second on SUN397, EuroSAT and Flowers102, where EM-Dirichlet, strong at B=64 but collapsing with large batches (Tab. 1), leads by 1.4, 4.3 and 0.4 points. Against StatA, GRAFT improves on every dataset; the margin is largest on the fine-grained vocabularies with many confusable classes, Stanford Cars (2.8), FGVC-Aircraft (2.6) and SUN397 (1.9), and smallest on the coarse ones, Caltech101 (0.6), EuroSAT (0.7) and UCF101 (0.8), as expected from a prior that pools displacements among semantically close classes. Table 4 extends the comparison to the ImageNet distribution shifts with the 1000-class vocabulary. The ranking is unchanged under the four shifts; TransCLIP remains far below the zero-shot model (29.1 against 46.2 on ImageNet-Sketch), StatA improves over zero-shot by 4.1 to

Table 1. Batch adaptation with a variable number of effective classes: top-1 accuracy (%) averaged over 1000 tasks and eleven datasets, CLIP ViT-B/16.

Method	**Very low** B=64	**Low** B=64	**Medium** B=64	**Medium** B=1000	**High** B=1000	**Very high** B=1000	**All** full
CLIP zero-shot [24]	65.2±0.6	65.2±0.5	65.2±0.4	65.2±0.2	65.2±0.2	65.2±0.1	65.2
MTA [35]	66.6±0.6	66.6±0.5	66.6±0.4	66.6±0.2	66.6±0.2	66.6±0.1	66.6
EM-Dirichlet [22]	68.5±0.7	70.3±0.6	67.5±0.5	64.4±0.4	45.3±0.6	33.6±0.7	29.5
ZLaP [16]	27.5±1.1	35.2±0.9	44.7±0.8	41.5±0.5	52.2±0.4	58.4±0.3	66.4
TransCLIP [36]	38.9±1.0	40.4±0.9	42.7±0.8	56.5±0.4	62.0±0.3	64.4±0.2	**70.3**
StatA [37]	70.4±0.4	69.3±0.3	67.4±0.3	69.7±0.2	69.8±0.2	69.0±0.2	69.9
MOON [14]	71.3±0.4	70.0±0.3	67.9±0.3	70.1±0.2	70.1±0.2	69.3±0.2	70.1
Unanchored mixture (α=λ=0, no π)	42.4±0.6	43.3±0.5	45.4±0.5	57.7±0.3	62.8±0.3	64.5±0.2	70.3
Presence only (π, no field)	67.4±0.5	66.7±0.4	65.3±0.4	66.9±0.2	66.6±0.2	65.7±0.2	66.0
Anchor-only shrinkage (λ=0)	70.1±0.4	69.1±0.3	67.3±0.3	69.6±0.2	69.7±0.2	69.0±0.2	69.9
Complete pooling (global shift)	69.6±0.4	68.8±0.3	66.8±0.3	68.8±0.2	68.6±0.2	67.6±0.2	68.3
GRAFT (fixed α, λ)	72.0±0.4	70.7±0.3	68.4±0.3	70.5±0.2	70.4±0.2	69.5±0.2	70.2
GRAFT + empirical Bayes	**72.1**±0.4	**70.8**±0.3	**68.5**±0.3	**70.6**±0.2	**70.5**±0.2	**69.5**±0.2	70.2

Table 2. Online adaptation: top-1 accuracy (%) over 100 runs and eleven datasets, batches of 128, CLIP ViT-B/16, for stream correlation $\gamma \in \{0.1, 0.01, 0.001\}$ and the separate-class stream.

Method	**Low**	**Medium**	**High**	**Separate**
CLIP zero-shot	65.2±0.3	65.2±0.3	65.2±0.3	65.2±0.3
TENT [33]	65.8±0.4	65.5±0.5	65.3±0.6	64.5±0.8
TDA [18]	**67.7**±0.4	67.1±0.5	66.8±0.6	66.6±0.7
DMN [41]	67.2±0.4	66.5±0.5	66.3±0.6	65.8±0.7
OGA [7]	67.5±0.5	69.1±0.5	69.6±0.6	69.1±0.7
StatA [37]	67.0±0.5	68.9±0.5	69.5±0.6	69.1±0.7
MOON [14]	67.4±0.5	69.4±0.5	69.9±0.5	69.4±0.6
GRAFT (transductive)	67.6±0.5	**70.1**±0.4	**70.8**±0.5	**70.2**±0.6
GRAFT (inductive)	67.2±0.5	69.5±0.4	70.3±0.5	69.8±0.6

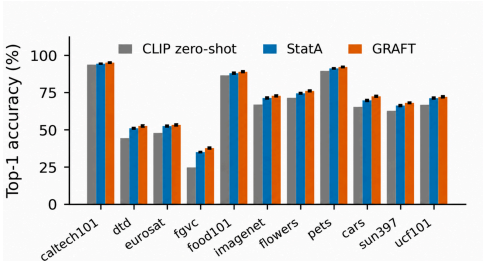


Figure 5. Per-dataset accuracy in the low setting for the zero-shot model, StatA and GRAFT.



Figure 6. Posterior mass assigned to classes absent from the batch versus the number of effective classes (ImageNet, B=64).

Table 4. ImageNet (IN) and its shifts V2, Sketch (S), A and R in the low setting, top-1 accuracy (%), CLIP ViT-B/16.

Method	**IN**	**V2**	**S**	**A**	**R**
CLIP zero-shot	66.7	60.8	46.2	47.8	74.0
TransCLIP	43.2	35.5	29.1	28.9	42.8
StatA	71.0	66.2	51.1	51.9	79.4
MOON	71.7	66.8	51.7	52.6	79.7
GRAFT	**72.3**	**68.3**	**53.1**	**54.3**	**81.9**

5.4 points, and the gain of GRAFT over StatA grows from 1.3 points on ImageNet to 2.0 to 2.5 points on the shifted sets, where the displacement of the image distribution from the text anchors, and with it the shared component that the graph propagates, is largest.

6.4. Qualitative results

Figure 7 shows test images from batches of the low setting with the ground truth and the predictions of the zero-shot model, StatA and GRAFT. The typical correction is a sample of a present class that the zero-shot model assigns to a semantically close class absent from the batch (an Egyptian cat for a tabby cat, a motor scooter for a moped); StatA recovers some of these samples when the present class has enough evidence to move its prototype, and GRAFT recovers the remaining ones because the absent neighbour is both removed through the presence model and displaced consistently with the present class. The failure case, a stone wall predicted as a cliff by all three methods, has no close neighbour in the batch. Figure 6 quantifies the mechanism on ImageNet batches of 64: with a single present class the zero-shot model assigns 31.0% of its posterior mass to classes that are not in the batch, StatA 11.5% and GRAFT 4.1%; with fifty present classes the values converge to 8.1%, 3.0% and 1.2%. The ratio between StatA and GRAFT is close to three at every K_{eff}, the effect of the presence term $\log \pi_k$ rather than of the field.

Table 3. Per-dataset top-1 accuracy (%) in the low setting (B=64, $K_{\text{eff}} \in [2, 10]$), CLIP ViT-B/16.

Method	IN	SUN	Airc.	Euro.	Cars	Food	Pets	Flow.	Calt.	DTD	UCF	Avg.
CLIP zero-shot [24]	66.7	62.6	24.8	47.8	65.3	86.1	89.1	71.3	93.3	44.4	66.7	65.3
MTA [35]	68.0	64.2	27.0	49.5	66.4	86.8	89.6	72.6	93.7	47.0	67.9	66.6
EM-Dirichlet [22]	70.2	**69.3**	35.1	**57.3**	70.3	88.0	91.0	**76.1**	94.1	50.6	71.4	70.3
ZLaP [16]	35.4	30.7	13.3	28.4	40.3	48.4	47.7	38.2	44.9	26.6	33.3	35.2
TransCLIP [36]	43.2	35.8	16.8	32.6	44.4	50.8	48.8	45.9	62.5	24.2	39.5	40.4
StatA [37]	71.0	66.0	35.0	52.3	69.6	87.4	90.7	74.2	94.0	50.9	71.1	69.3
MOON [14]	71.7	66.9	36.3	52.6	71.0	87.9	91.2	75.0	94.3	51.6	71.5	70.0
GRAFT	**72.3**	67.9	**37.6**	53.0	**72.4**	**88.5**	**91.6**	75.7	**94.6**	**52.3**	**71.9**	**70.7**

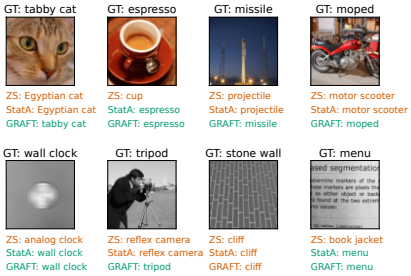


Figure 7. Qualitative predictions in the low setting: ground truth (GT) and the predictions of the zero-shot model (ZS), StatA and GRAFT; correct predictions in green, wrong ones in red.

Table 5. Component ablation, top-1 accuracy (%): π presence, α anchor, λ graph coupling, EB empirical Bayes, W_s sample graph.

π	α	λ	EB	W_s	Low (B=64)	All
zero-shot					65.2	65.2
–	–	–	–	✓	43.3±0.5	70.3
✓	–	–	–	✓	55.8±0.5	70.3
–	✓	–	–	✓	69.1±0.3	69.9
✓	✓	–	–	✓	69.6±0.3	69.9
–	✓	✓	–	✓	70.2±0.3	70.2
✓	✓	✓	–	–	70.4±0.3	68.7
✓	✓	✓	–	✓	70.7±0.3	70.2
✓	✓	✓	✓	✓	**70.8**±0.3	70.2

7. Ablation Studies

Components. Table 5 adds the components one at a time, in the low setting and on the full test set. Without anchor, coupling or presence the mixture reaches 43.3 in the low setting, 22 points below the zero-shot model, but 70.3 on the full set, where every class is well observed. The presence model alone raises the low-setting result to 55.8 but cannot prevent the present classes from drifting; the anchor term alone raises it to 69.1, and the two together to 69.6, the anchor-only regime of StatA and MOON. The graph coupling adds 1.1 points on top (70.7), the margin that distinguishes GRAFT from the anchored methods, and empirical Bayes a further 0.1. The inherited sample graph contributes 0.3 points in the low setting but 1.5 on the full set, where neighbouring samples carry most of the structure; conversely, the anchor and the coupling change the full-set result by at most 0.4 points, consistent with the special-case analysis of Sec. 4.2.

Prior and graph. We vary α, λ and the number of neighbours m, and replace the text kNN graph by a complete graph and by a degree-matched random graph (table and sensitivity curves in the supplementary material). Accuracy in the low setting varies by at most 0.6 points for λ between 1 and 16 and by 0.3 points for m between 3 and 10, so neither is a sensitive hyper-parameter; removing the anchor ($\alpha = 0$) costs 1.7 points, removing the coupling ($\lambda = 0$) 1.6, and very large values ($\lambda = 64$, $\alpha = 16$) cost 0.8 and 1.1 points by over-smoothing or over-shrinking the field. The random graph (69.2) performs like no coupling (69.1) and the complete graph (68.8) is below both, so pooling helps only when it follows the semantic neighbourhoods. This is the control that the smoothness hypothesis H1 of Sec. 4.2 requires. H1 itself is tested with labels, independently of any method (plot in the supplementary material): the Dirichlet energy of the oracle displacement field on the text graph, relative to its mean on twenty degree-matched random graphs, lies between 0.25 and 0.75 on all eleven datasets, and the datasets with the highest ratios are among those with the smallest margins in Tab. 3. Likelihood and optimisation variants change the result by at most 0.5 points, and a batch of 64 ImageNet images takes 9 ms against 7 ms for StatA on one GPU (supplementary material).

8. Conclusion

We traced the collapse of unlabeled VLM adaptation on realistic batches to independent per-class parameters and replaced them by a displacement field over the vocabulary with a Gaussian Markov random field prior on a semantic class graph and an estimated class prior. GRAFT is closed form, has anchor-only shrinkage and a global shift as extreme settings, and improves over anchored transduction where few classes are present.

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LLM Usage Statement

In accordance with the CVPR 2027 call for papers, we disclose that a large language model based coding agent was used in the preparation of this submission: it assisted in surveying the literature, drafting the manuscript text, implementing the released code and producing the figure scripts, under the direction and review of the authors. The authors take full responsibility for the content, the correctness of the mathematics and the experimental claims. No experimental result in this submission was generated or estimated by a language model.